

THE CLASSICAL PERIOD

500 B.C. to A.D. 500

rapid change was taking place in Greek society

agrarian monarchy to a commercial industrial democracy

the moral focus was the duty of the individual as a citizen

Ethics was regarded as the spiritual counterpart of medicine, its function being to provide care of

the soul as medicine provides care of the body

Their general view was that good and evil are matters of personal decision or social agreement.

Some of the more liberal Sophists suggested that all morality was a matter of convenience only.

Socrates (469-399 B.C.)

was the dominant figure of the classical period

he is generally regarded as the father of Western philosophy.

Like the Sophists, Socrates rejected the idea that tradition justifies conduct

he believed that morality is more than a matter of personal choice or convenience.

It is possible, he believed,

to develop a universal set of ethical principles to guide conduct, and

the key to doing so is human reason,

the careful examination of beliefs and actions and

the logic that underlies them.

His focus was on self-knowledge.

"The unexamined life is not worth living."

Socrates is best known for his philosophical method.

Rather than teach directly,

he conversed with others,

asking basic questions about important matters (for example, What is justice? What is virtue?) and

then examining the answers people gave,

exposing vagueness and inconsistency,

clarifying problems, and

pointing the way to further inquiry.

He was also the first to probe the relationships between facts and values an ethical issue that virtually all ethicists since his time have grappled with.

Plato (427-347 B.C.)

Plato (427-347 B.C.) built upon the teaching of Socrates.

Plato's writings are in dialogue form, and Socrates is the main character in most of them.

Plato's greatest work is The Republic.

He shared Socrates' view that the life of reason is the happiest and best life.

For both men sense perception and bodily pleasure were less desirable than intellectual pursuits, but the two men differed somewhat in their emphases.

Socrates apparently believed in

enjoying himself at all levels of experience in a balanced way.

nothing should be done to excess,

that moderation should be the rule in everything.

Plato's view was more extreme.

His denial of bodily pleasure and sense perception made him a model for later religious and idealistic ethicists.

Plato's central notion was that

the real world is not the world that our senses perceive,
but the world of ideas.

The concrete reality that surrounds us is merely an imperfect reflection of the world of abstract ideas or "forms," which are timeless and unchanging.

The most important reality in this world of ideas is the idea of the Good.

Goodness for Plato was a central fact about the universe.

Thus the main goal of his ethical system is to gain a vision of the Good.

Aristotle (384-322 B.C.)

Aristotle (384-322 B.C.) was a student of Plato and followed his and Socrates' philosophical tradition.

Nevertheless, Aristotle's emphasis was quite different from Plato's.

Aristotle authored as many as four hundred works on a variety of subjects from theoretical and practical science to politics, rhetoric, logic, and ethics.

His Nichomachean Ethics is the first systematic treatment of ethics in Western civilization.

Aristotle disputed Plato's theory of forms.

There is no world of separate, abstract forms corresponding to the concrete things we perceive, he reasoned.

Form cannot exist apart from particular objects, or as he put it, "No form without matter, no matter without form."

Accordingly, Aristotle rejected the notion that the Good exists independently of daily experience and human personality.

In his view moral principles exist in the daily activities of human life and can be discovered by examining those activities.

Happiness is to be attained by developing one's potential for a life of reason.

The life of reason has two aims

the pursuit of truth through reflection and
understanding and the pursuit of virtue through intelligent conduct.

Virtue, for Aristotle, represented a midpoint between extremes of excess and defect.

(Some actions, however, like murder or theft are bad in themselves and therefore having no midpoint.)

The Cyrenaics, the Epicureans emphasized human feelings and desires and taught that the measure of an action is the amount of pleasure it brings. (This view is known as hedonism.)

Epicurus made a sharp distinction between "natural" pleasures, such as peace of mind and the absence of hunger, and "unnatural" pleasures, such as greed and lust, approving only the former.

In contrast to the Cyrenaics and Epicureans were the Cynics and then the Stoics.

These groups stressed overcoming feelings and desires and serving the demands of duty as revealed by reason.

THE MEDIEVAL PERIOD (500 to 1500)

Its intellectual and social context was Christianity

Medieval ethics combined the classical emphasis on human reason with the idea of obedience to God's will.

Saint Augustine (354-430)

The first was Saint Augustine (354-430). In his numerous works, notable among them *Confessions* and *The City of God*,

Augustine made Plato's philosophy the basis of Christian ethics.

Augustine's system was two-sided -

- the life of reason leads to temporal well-being

- faith leads to salvation and eternal happiness.

(Since this life is only a preparation for the afterlife, no real happiness is possible here.)

The concept of Good for Augustine was similarly two-sided:

- the natural earthly side and

- the supernatural, other-worldly side, with the super-natural dominating.

Augustine viewed man as having fallen from God's grace through original sin but retaining free will and responsibility for his actions.

Virtue is possible, but to have real value it must be prompted by faith.

Saint Thomas Aquinas (1225-1274).

Aquinas rediscovered Aristotle, Christianized his philosophy, and made it the basis of the philosophical outlook of the later middle ages.

Aquinas's greatest works were *Summa Theologica* and *Summa Contra Gentes*.

Aquinas aimed at demonstrating the harmony between Aristotle and Christianity.

He taught that ethics has two dimensions-

- the natural and

- the theological.

Natural ethics, as detailed by Aristotle, consists of the development of reason and practice in living morally and leads to earthly happiness.

Theological ethics consists of achieving the virtues of faith, hope, and charity through God's grace and leads to eternal life with God.

Aquinas believed that the Natural Law-the divine law as written in the heart of man can be discovered by reason and cultivated by conscience.

By allowing people to turn to secular knowledge without guilt,

Aquinas prepared the way for the emergence of a more scientific view of humanity and of ethics.

THE MODERN PERIOD (1500 to the present.)

The Protestant Reformation challenged the supremacy of the "Roman Catholic church and introduced the idea of each person interpreting the Bible for himself or herself.

The impact of this idea was increased with the invention of the printing press and the shift from Latin to the vernacular.

Finally, and perhaps most importantly, the work of scientists like Copernicus, Galileo, and Harvey turned the attention of philosophers from theological to scientific explanations.

Thomas Hobbes (1588-1679)

Thomas Hobbes (1588-1679) was the first thinker to systematically approach ethics from a scientific viewpoint.

Hobbes, whose major philosophical work was *Leviathan*, argued that the notion of mechanistic materialism that was central to the physics of the time applied to ethics as well.

He attempted to demonstrate that in humanity's natural (primitive) condition the rule of self-preservation produces a morality based on self-interest.

In that condition, Hobbes reasoned, the concepts of right and wrong do not exist; they only begin to exist when civil society is formed.

The only way civil society can control the pursuit of individuals' self-interest and the conflict that such pursuit inevitably generates is to have everyone give allegiance to the sovereign.

Hobbes believed in the Golden Rule, but he also believed that people could not be trusted to practice it.

With too much at stake for each person to follow his or her conscience, the sovereign power—whether "monarch or assembly"—must ensure "the safety of the people."

There was strong reaction against Hobbes' ethical perspective

human beings are not at the mercy of their drive for self-preservation.

They have a moral faculty, a specific intellectual guide that enables them to tell right from wrong.

Several different ways of describing that moral faculty were advanced.

Some described it as intuition, others as a moral sense (a natural affection for virtue), and still others as conscience.

David Hume (1711-1766)

In the early eighteenth-century David Hume (1711-1766) proposed an ethical theory that was in some ways reminiscent of John Locke's theory in the previous century.

Locke (1632-1704) had argued that pleasure is the standard of moral judgment.

Sensations of pleasure or pain he reasoned, lead us to reflect and form views of justice and goodness and thus develop a system of moral judgment.

Hume believed that the standard of moral judgment is two-sided.

One part of it is objective—the consequences of the action in question.

The other part, the dominant part, is subjective—a feeling of pleasure.

(Like most hedonistic philosophers Hume did not approve of doing anything we wish. He believed that every human being possesses a tendency to be more pleased by benefiting others than by being selfish and self-indulgent.)

In Hume's theory reason alone cannot answer moral questions.

But a "moral sentiment" that chooses what is useful or pleasant can and does do so.

Immanuel Kant(1724-1804)

Immanuel Kant is one of the most influential philosophers in the history of ethics, and his work is considered the very basis of modern ethical discussion.

His ethical theory is known as **Deontology** — which comes from the Greek word *deon*, meaning duty.

The central idea of Kant's ethics is that morality must be grounded in **pure reason**, not in emotion or feeling.

For Kant, acting morally has nothing to do with how you feel about a situation — it is entirely about whether your action follows a rational, logical principle.

At the heart of Kant's moral philosophy is the concept of "**Good Will.**"

Kant believed that the only thing in the world that is unconditionally good is a good will — meaning the intention to do what is right simply because it is right, not for any other personal benefit or emotional reason.

This good will serves as the foundational imperative of all moral action.

If a person does the right thing for the wrong reason — such as to impress others or to avoid punishment — Kant would say that action has no genuine moral worth.

Kant also believed that **happiness is achieved through reason**, not through the pursuit of pleasure or the satisfaction of desires.

In a civilized society, human beings are regarded as rational beings — and it is precisely this rationality that gives them their dignity and moral status.

Finally, Kant argued that natural law operates as what he called **categorical imperativeness** — meaning there are absolute, universal moral commands that every rational person is obligated to follow, in every situation, without exception.

Kant's Categorical Imperative

The Categorical Imperative is Kant's supreme principle of morality —

a universal rule derived entirely from reason that tells us how we must act in all circumstances.

It is "categorical" because it is unconditional — it does not say "do this if you want that outcome." It simply says: this is what you must do, regardless of consequences.

Kant expressed this imperative in **three** key formulations.

The first is to **act according to universal law** — before taking any action, you must ask yourself: "What if everyone in the world did exactly what I am about to do?" If the result would be a world that functions well, then the action is permissible. If the world would collapse into chaos or contradiction, then the action is wrong.

The second formulation is to always **treat humans as ends, not merely as means** — meaning you must never use another person simply as a tool or instrument for achieving your own goals. Every human being has intrinsic worth and dignity that must always be respected.

The third formulation is that **duty gives moral worth** to an action — the rightness or wrongness of what you do is not determined by the outcome it produces, but by whether it was done out of a sense of moral duty. An act done from duty is morally worthy; the same act done from selfishness or convenience is not.

John Stuart Mill (1806–1873)

John Stuart Mill represents a fundamentally different approach to ethics from Kant.

Where Kant focused on duty and intention, Mill focused entirely on **consequences**.

His ethical theory is called **Utilitarianism**, which falls under the broader category of **Teleology** — meaning ethics that is concerned with ends and outcomes rather than rules and duties.

Mill was deeply inspired by two earlier thinkers —

his father **James Mill** and the philosopher **Jeremy Bentham** — who had laid the early foundations of utilitarian thought.

The core principle of Mill's Utilitarianism is straightforward:

the rightness or wrongness of any action is determined entirely by its **consequences**.

An action is morally right if it produces good outcomes, and
morally wrong if it produces bad ones.

This is why Mill's ethics is also described as **Hedonist** — because the standard by which he measures good and bad consequences is **pleasure and happiness**.

For Mill, the ultimate goal of any moral action should be to produce the greatest amount of happiness for the greatest number of people.

Pleasure and happiness are not just desirable — they are the fundamental standard of all moral judgment.

Mill's Two Types of Motivation

Mill also explored the question of *why* people act morally — what motivates human beings to do good.

He identified two distinct types of motivation.

The first is **external motivation** — acting morally because of outside pressures, such as wanting to earn a friend's approval or fearing punishment, whether from society or from God's wrath. This type of motivation comes from outside the individual and is driven by reward and fear.

The second and deeper type is **internal motivation** — which Mill described as an inner feeling of humanity. This is a genuine, inward sense of moral obligation that a person feels not because of outside pressure, but because they have developed a sincere concern for other people and for doing what is right.

Importantly, Mill believed that this internal motivation is not something people are simply born with — it is something we **learn from society**.

Through our social experiences, relationships, education, and cultural life, we gradually develop this inner moral sense that guides us to act well even when no one is watching and no external reward or punishment is involved.

THE CLASSICAL PERIOD

Socrates — "The Unexamined Life is Not Worth Living"

Socrates used questioning to expose vague or inconsistent thinking — this is **code review and debugging** in CSE.

Real CSE Example: When a developer writes `if (x = 5)` instead of `if (x == 5)`, a good code reviewer (like Socrates) doesn't just say "it's wrong" — they ask "*what were you intending here?*" and expose the logical inconsistency. Tools like **ESLint**, **SonarQube**, and **GitHub PRs** are literally Socratic methods built into software engineering pipelines.

Plato — World of Ideas (Abstract Forms)

Plato believed the real world is made of **abstract, perfect, unchanging ideas**, not physical things.

Real CSE Example: This is **Object-Oriented Programming (OOP) and Abstraction**. A `Car` class is Plato's *Form of Car* — it's the perfect abstract template. Every actual car object (`myCar`, `yourCar`) is just an imperfect physical instance of that ideal form. Interfaces in Java/C# are pure Platonic forms — they define what something *must be*, with no physical implementation.

Aristotle — "No Form Without Matter"

Aristotle rejected abstract forms and said goodness is found **in actual daily activities** — virtue is a *midpoint* between extremes.

Real CSE Example: This is **algorithmic balance and optimization**. In system design, Aristotle's "golden mean" is literally the engineer's trade-off:

- Too little memory → crashes (defect)
- Too much memory allocated → wasteful (excess)
- Right amount → virtue (optimal design)

This also maps to **Agile development** — Aristotle wouldn't write all docs upfront (Waterfall excess) or zero docs (Kanban defect). He'd write *just enough* documentation.

Epicureans vs. Stoics — Hedonism vs. Duty

Epicureans maximized pleasure; Stoics followed reason and duty regardless of feeling.

Real CSE Example:

- **Hedonist developers** write whatever code *feels* fast to ship (tech debt, spaghetti code)
- **Stoic developers** follow **clean code principles**, SOLID principles, and proper documentation even when deadlines are painful

The eternal battle in every tech company: "*Let's just ship it*" (Epicurean) vs. "*Let's write unit tests first*" (Stoic).

THE MEDIEVAL PERIOD

Aquinas — Natural Law Discoverable by Reason

Aquinas said divine law is written in nature and can be found through **rational examination**.

Real CSE Example: This is **algorithm discovery and mathematical proofs** — the laws of computation (like P vs NP, Turing completeness, Big-O complexity) *exist in nature*, and computer scientists discover them through pure reason, just as Aquinas described. Dijkstra didn't *invent* the shortest path — he *discovered* a law already present in the mathematics of graphs.

THE MODERN PERIOD

Hobbes — Self-Interest Without Rules Creates Chaos

Hobbes said without a sovereign enforcing rules, everyone acts in self-interest and society collapses.

Real CSE Example: This is **cybersecurity and internet governance**. Without protocols, firewalls, and laws:

- The internet would be a Hobbesian "war of all against all" — DDoS attacks, data theft, ransomware, spam
 - **TCP/IP, HTTPS, OAuth, and JWT** are the "sovereign" — the agreed-upon rules that make cooperation possible
 - Without authentication systems (the sovereign), every API would be exploited purely out of self-interest
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Hume — Moral Judgment is Both Objective + Subjective

Hume said morality has an objective part (consequences) AND a subjective part (human feeling/sentiment).

Real CSE Example: This is **UI/UX design ethics**. A dark pattern (like hiding the "unsubscribe" button) may be *objectively* functional — it works, it's technically correct. But the *subjective moral sentiment* of users judges it as manipulative and wrong. **Hume would say:** measure the consequences (user trust drops, churn rises) AND the feeling (users feel tricked). This is exactly how companies like Apple frame their **Human Interface Guidelines** — balancing objective functionality with subjective user dignity.

Kant — Deontology & Categorical Imperative (Most Relevant to CSE Ethics)

Kant said: act only according to rules you'd want *everyone* to follow. Treat people as ends, never merely as means.

Real CSE Example 1 — AI Ethics: If an AI model scrapes data without user consent, Kant's first formulation asks: "*What if every company did this?*" → The internet becomes untrustworthy. This is exactly why **GDPR** and **AI Ethics frameworks** exist. Kant's categorical imperative is literally the foundation of privacy law.

Real CSE Example 2 — Software Piracy: A developer asks: "*Should I use this cracked library?*" Kant says: if everyone did this, software companies couldn't survive → no more software → contradiction. Therefore it is morally wrong, regardless of whether you get caught.

Real CSE Example 3 — Treating Users as Ends: Facebook's early business model used users as *means* (selling their data). Kant would condemn this. Modern **Privacy-by-Design** principles in software engineering directly implement Kant's second formulation — users have intrinsic dignity and must not be exploited as data sources.

Mill — Utilitarianism & Consequentialism (The Engineer's Go-To Ethics)

Mill said: an action is right if it produces the **greatest happiness for the greatest number**.

Real CSE Example 1 — Search Algorithms: Google's PageRank was utilitarian — rank results that give the *most benefit to the most users*. Every recommendation system (Netflix, YouTube, Spotify) is a utilitarian machine: maximize aggregate satisfaction.

Real CSE Example 2 — Trolley Problem in Self-Driving Cars: If an autonomous vehicle must choose between hitting one pedestrian or five, it's a pure Millian calculation. Tesla, Waymo, and other AV companies literally had to encode utilitarian decisions into their systems — and faced enormous ethical backlash for it.

Real CSE Example 3 — Internal vs External Motivation: Mill's internal motivation maps to **developer intrinsic motivation** — senior engineers who write clean code not because a manager is watching, but because they've internalized professional values. This is what separates a **craftsman developer** from a **ticket-closer developer**.

Summary Table

Philosopher	Core Idea	CSE Equivalent
Socrates	Question everything	Code review, debugging, testing
Plato	Abstract ideal forms	OOP, interfaces, abstraction
Aristotle	Virtue = balance	System design trade-offs, Agile
Hobbes	Rules prevent chaos	Cybersecurity, protocols, governance
Hume	Objective + subjective judgment	UX ethics, dark patterns
Kant	Duty-based universal rules	GDPR, AI ethics, privacy-by-design
Mill	Greatest good for greatest number	Recommendation systems, AV ethics

The entire history of ethics your document covers is **actively being re-enacted in CSE today** — every AI model trained, every system designed, and every policy written is a new chapter in this same philosophical story.